

TAIPAQ, Kazakhstan

WMO: 354060

Lat: 49.050N

Lon: 51.867E

Elev: 7

StdP: 14.69

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.5	-8.6	-19.8	1.9	-13.9	-14.8	2.5	-7.8	27.3	18.9	20.6	17.3	3.3	50	0.507

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.0	99.7	68.6	96.6	67.5	93.5	66.3	71.0	93.0	69.1	91.4	67.7	89.7	6.2	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.4	90.6	78.3	61.3	81.1	77.5	59.1	74.7	77.5	34.8	92.9	33.2	91.6	32.0	89.7	82.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	26.8	18.3	14.5	DB	-18.1	103.7	8.2	3.1	-24.0	106.0	-28.8	107.8	-33.5	109.5	-39.4	111.8
(5)				WB	-18.0	73.3	8.1	3.3	-23.8	75.6	-28.5	77.6	-33.1	79.4	-38.9	81.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.8	14.7	16.7	31.9	51.2	64.7	74.0	78.5	76.1	62.7	47.2	32.2	21.2
	DBStd	24.69	12.37	12.96	10.59	9.18	8.66	7.84	6.41	7.93	8.33	8.68	10.03	10.79
	HDD50	4286	1095	932	564	94	5	0	0	0	6	160	536	894
	HDD65	7537	1560	1352	1027	421	114	16	1	11	138	554	984	1359
	CDD50	3468	0	0	3	130	461	719	883	809	389	72	2	0
	CDD65	1240	0	0	0	7	105	284	419	354	70	1	0	0
	CDH74	15636	0	0	0	111	1280	3572	5304	4482	861	25	1	0
	CDH80	7815	0	0	0	20	519	1807	2787	2367	312	3	0	0
Wind	WSAvg	5.4	5.4	5.8	6.2	6.3	5.6	5.2	4.7	4.8	4.7	5.0	4.8	5.8
Precipitation	PrecAvg	7.9	0.6	0.6	0.6	0.7	0.6	0.9	0.7	0.8	0.6	0.8	0.9	0.8
	PrecMax	13.8	1.2	1.6	2.4	2.9	1.9	3.8	4.7	3.5	1.9	2.8	3.5	2.3
	PrecMin	4.2	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.0
	PrecStd	2.3	0.3	0.5	0.5	0.6	0.5	0.9	0.8	0.8	0.5	0.6	0.7	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.1	40.8	62.5	83.2	94.0	101.7	102.9	102.6	94.2	76.8	58.1	42.7
		MCWB	35.8	36.1	49.2	61.0	65.6	68.7	69.5	69.4	66.3	58.1	50.0	38.0
	2%	DB	34.9	36.8	55.5	77.2	89.9	97.6	99.4	99.5	87.4	71.9	52.4	39.0
		MCWB	33.3	33.6	45.9	58.1	64.2	67.4	68.7	68.4	63.2	55.4	46.4	37.1
	5%	DB	33.2	34.7	49.8	72.6	86.4	94.0	96.3	95.9	82.9	66.9	48.3	36.3
		MCWB	31.6	32.3	42.2	56.3	62.8	65.8	67.7	67.4	61.0	52.7	44.4	33.9
	10%	DB	31.2	33.1	45.0	67.8	81.7	90.3	92.9	92.2	78.7	61.9	44.6	33.6
		MCWB	29.7	31.1	39.4	53.5	61.1	64.7	66.6	66.3	59.4	50.2	41.1	31.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.9	36.9	50.8	62.8	68.4	72.4	75.0	74.8	68.8	60.8	51.9	40.4
		MCDB	37.7	39.6	59.8	79.4	89.4	92.9	93.1	90.7	88.3	73.1	57.2	41.8
	2%	WB	33.6	34.2	46.5	60.0	65.6	69.0	70.9	71.0	64.5	56.5	47.7	37.3
		MCDB	34.8	36.1	54.6	75.5	85.6	91.9	93.5	92.2	84.9	68.7	51.2	38.9
	5%	WB	31.9	32.8	43.0	57.0	63.6	67.4	69.1	68.9	61.9	53.8	44.6	34.2
		MCDB	33.2	34.5	49.0	70.2	82.6	90.0	91.2	91.7	80.5	64.9	48.2	35.7
	10%	WB	29.8	31.3	40.0	54.3	61.9	65.8	67.7	67.2	59.9	51.2	41.4	32.1
		MCDB	31.3	33.2	44.8	66.4	79.8	87.0	88.7	89.6	76.7	60.3	44.4	33.6
Mean Daily Temperature Range	5% DB	MDBR	12.2	14.3	14.8	20.9	21.8	22.4	22.0	23.4	22.9	19.3	12.5	10.5
		MCDBR	8.2	11.7	21.2	26.8	27.1	26.8	25.6	27.2	29.2	27.3	17.4	10.0
		MCWBR	7.5	9.7	13.2	13.9	11.1	9.7	9.2	9.4	13.1	15.3	12.6	8.5
	5% WB	MCDBR	7.7	10.1	19.2	24.7	25.1	24.3	23.2	25.0	27.1	24.6	15.8	8.7
	MCWBR	7.5	8.9	12.8	13.6	11.7	9.8	10.0	10.5	13.4	14.9	12.3	8.1	
Clear-Sky Solar Irradiance	taub	0.331	0.351	0.407	0.462	0.440	0.422	0.421	0.434	0.387	0.369	0.348	0.333	
	taud	2.077	2.033	2.066	1.993	2.134	2.253	2.269	2.205	2.293	2.264	2.220	2.130	
	Ebn at Noon	214	242	249	248	261	267	265	255	256	237	210	196	
	Edh at Noon	28	37	43	52	47	42	41	41	34	30	25	24	
All-Sky Solar Radiation	RadAvg	369	716	1066	1505	1945	2145	2078	1808	1331	784	380	245	
	RadStd	72	116	130	108	149	167	115	99	105	88	48	54	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	+2.38	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page